

Geometry Unit 8 Assessment Guide

General Information	Unit 8 explores quadrilaterals. Throughout the unit, students are observing characteristics of the different types of quadrilaterals so that they gain an understanding of which characteristics are properties that can be used to categorize the quadrilateral. Students then use these properties to solve problems. The unit includes kites as an extension of the benchmarks. Question 1 is the only question on the unit assessment where a kite is used. Often students stumble when asked to restate properties of given quadrilaterals. The draft item specifications state that “items will not require the student to identify a reason using/recognizing the formal name of a theorem/postulate/relationship unless referenced in the benchmark or clarifications.” It also states, “items will not assess postulates, relationships, and theorems beyond Clarification 1.” The relationships for parallelograms are opposite sides are congruent, consecutive angles are supplementary, opposite angles are congruent, the diagonals of a parallelogram bisect each other, and rectangles are parallelograms with congruent diagonals. The relationships for trapezoids are the Trapezoid Midsegment Theorem and for isosceles trapezoids: base angles are congruent, opposite angles are supplementary and diagonals are congruent. The Geometry B1G-M repeatedly states students should be aware of the B.E.S.T. definition of a trapezoid. This definition is applied in question 1. Teachers may wish to observe whether students are overlooking the inclusive definition. If teachers observe students are confused, then it may be helpful to revisit the Warm-Up in lesson 5. Question 2 requires the students to give a reason that is prior knowledge. Teachers may wish to monitor students to determine whether students struggle with this prior knowledge. Teachers may benefit from knowing that if this proof was on the statewide assessment that the last reason would not be a requirement for a proof that was assessing MA.912.GR.1.4. An item on the statewide large-scale assessment must show knowledge of just the benchmark it is aligned to. This is a requirement to ensure the assessment’s validity.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question		1							Benchmark(s)	MA.912.GR.1.4, MA.912.GR.1.5
									Recommended Scoring (34 Total Points)	Consider giving students 2 points for each correct answer and to deduct 1 point for each incorrect selection.

Question	2	Benchmark(s)	MA.912.GR.1.4
Parallelogram WBHN Given	$\overline{WB} \cong \overline{NH}$ Definition of parallelogram	Recommended Scoring (15 Total Points)	Consider giving students 5 points for each correct reason in the proof.
		Additional Notes	It is important to notice that the last reason in the proof is CPCTC which is prior knowledge. The other two reasons in the proof are showing knowledge of the benchmark being assessed whereas the reason for the last step is not.
$\angle BWK \cong \angle RHN$ Definition of parallelogram	$\angle WBK \cong \angle HNR$ Given $\triangle WBK \cong \triangle HNR$ ASA $\overline{KB} \cong \overline{NR}$ CPCTC		

Question	3	Benchmark(s)	MA.912.GR.1.5
	2.3	Recommended Scoring (9 Total Points)	Consider giving students 3 points for correctly setting up the relationship between the bases and the midsegment, 3 points for partial credit on solving the equation, and 3 points for the correct answer.
Question	4	Benchmark(s)	MA.912.GR.1.4
	$2(6x - 7.7) = 221$	Recommended Scoring (5 Total Points)	Consider giving students 5 points for correctly setting up the relationship.
Question	5	Benchmark(s)	MA.912.GR.1.4, MA.912.GR.1.5
	☐ Base angles of Quadrilateral T are congruent. ☐ Base angles of Quadrilateral Z are congruent. ☐ Opposite angles of Quadrilateral C are congruent. ☐ Opposite angles of Quadrilateral L are congruent. ☐ Consecutive angles of Quadrilateral Z are congruent ☐ Consecutive angles of Quadrilateral L are congruent. ☐ Opposite angles of Quadrilateral C are supplementary. ☐ Opposite angles of Quadrilateral T are supplementary.	Recommended Scoring (20 Total Points)	Consider giving students 5 points for each correct answer. Consider deducting 5 points for each incorrect choice.

Question	6	Benchmark(s)	MA.912.GR.1.5
work $12^2 + 20^2 = DB^2$ $DB \approx 15.62$ $16^2 + 8.72^2 = KR^2$ $KR \approx 18.22$ $6.83 + 18.22 = YR$ $25.05 = YR$		Recommended Scoring (17 Total Points)	Consider giving students 3 points for using the Pythagorean theorem to find RK, 3 points for finding RY, 3 points for using the Pythagorean theorem to find DB, 4 points for correctly selecting “is not” in the first statement, and 4 points for correctly stating “does not” in the second statement. Teachers may wish to consider that students whose work does not justify their answer choices in the statements should not receive the points for their choice.
Complete the statements. Jaxson’s paper airplane ☐ is ☒ is not an isosceles trapezoid. Jaxson’s paper airplane ☐ does ☒ does not meet the regulations.		Additional Notes	Students may be able to determine if the trapezoid is isosceles using a point on DR that is the altitude from Y. This makes an assumption that should not be made. Students will still need to find the diagonals. This method cannot be avoided but the diagram makes it more difficult, and an assumption is made by the student.